Beyond Shot and Group Counts: Support Allocation in Low-Shot Landmark Gesture Classification

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***Abstract*— Few-shot evaluations usually report the number of support examples and acquisition groups, but these quantities do not determine how support records are allocated across groups. We study this omitted allocation variable in known-class landmark gesture classification using released HaGRIDv2 landmarks and 18 command classes. The primary pre-exposure endpoint compares one-group and five-group support at $K = 5$; a post-test extension compares all seven allocations while matching each record's marginal inclusion probability. At fixed $U = 2$, redistributing four plus one records into three plus two improves coordinate and mixed-geometry ridge recall by 1.93 and 1.73 percentage points, with approximate simultaneous 95% intervals [1.33, 2.52] and [1.14, 2.33]. The same fixed-group-count effect appears under a post-test RBF SVM analysis, while prototype controls show that reduced support dispersion does not imply higher recognition recall. These results are conditional on the fixed landmark population and query protocol, and show that support allocation q should be reported and matched in support-set comparisons.**

I. Introduction

Five shots from two acquisition groups can mean four plus one records or three plus two (Fig. 1). The same shot count and group count therefore leave the support contribution vector unspecified. That vector can change recognition even when the included-record marginals are matched.

Few-shot comparisons depend on representations and support-set classifiers [5], [25]. Image-uniform and group-uniform sampling reweight differently sized groups, so they do not isolate allocation. We compare all seven five-record partitions from a common pool, randomize group positions, and match record-inclusion probabilities. The expected class prototype is preserved while the support covariance and decision boundary can change.

The primary endpoint establishes the one-group versus five-group contrast. The post-test extension then tests the allocation axis directly: at $K = 5, U = 2$, the ridge gains are 1.73–1.93 pp, with simultaneous lower bounds of 1.14–1.33 pp; the corresponding three-group effects are smaller. Covariance and classifier controls separate support dispersion from recognition recall.

We use 18 known HaGRIDv2 classes, released 2-D landmarks, coordinate and mixed features, prototypes, ridge, and RBF SVM. The original endpoint remains primary, while allocation and SVM analyses are post-test extensions on the same population. The resulting reporting rule is direct: support-set comparisons should report K, U, and q, and models in one comparison should use the same allocation.

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II. Related Work

Task and learner controls. Matching Networks and Prototypical Networks classify queries from labeled support examples [28], [24]; MAML adapts model parameters [10], while ANIL studies feature reuse and head adaptation [20]. SimpleShot emphasizes feature normalization and nearest-neighbor rules [29]; MetaOptNet differentiates through a convex support-set learner [15]; Meta-Baseline combines supervised pretraining and episodic optimization [6]. These works separate representation and adaptation choices. Meta-Dataset additionally varies ways, shots, and class balance [26]. Our fixed landmark features isolate a support-sampling variable within known classes.

Acquisition and representation controls. HaGRID and HaGRIDv2 provide gesture images and acquisition metadata [13], [19]. MediaPipe defines reusable perception pipelines [17], and MediaPipe Hands combines palm detection with landmark estimation [31]. WLASL compares appearance and pose-based sign recognition [16]. MS-ASL and the AUTSL challenge evaluate unseen signers [12], [23]. Signer-disjoint partitions control exposure across splits; they leave within-support contribution counts as a separate design choice. Here `user_id` denotes an acquisition group, not verified natural-person identity. Matched landmark representations [18] and five-class, K-support, 15-query sign episodes [4] control other protocol axes; their inspected descriptions leave contributor counts unspecified.

Support construction controls. Fixed support-pool reuse across meta-training tasks [22] and exemplar selection from three candidates per HaGRID class [11] are direct precedents for changing support construction. The latter describes query-dependent retrieval in its method but filename-order selection in evaluation. VariFace compares 0.5M images allocated as 10K identities times 50 images or 25K times 20 [30]. Its identities are target classes. Our groups lie within

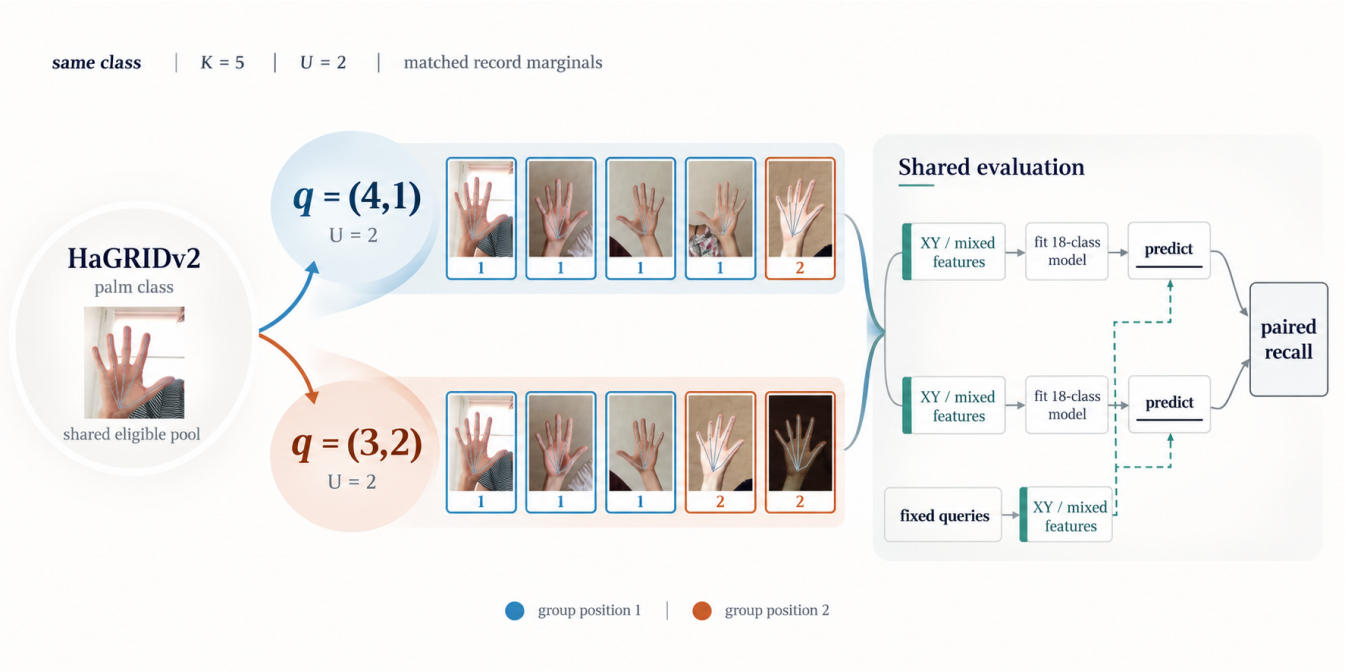


Fig. 1. Support allocation at $K = 5, U = 2$, illustrated with hand images and landmark overlays. The two branches differ only in q: $(4, 1)$ versus $(3, 2)$. Blue and orange borders mark randomized ordered group positions, not identities. Both branches use the same feature rules, 18-class fitting protocol, and fixed queries; classifiers use landmark features rather than RGB crops. The image is an illustrative schematic of the protocol and does not represent a reported experimental draw.

each fixed gesture class: we hold K and U fixed and match record marginals while changing their contribution vector.

Comparison and inference controls. Benchmark variance [2], model-selection bias [3], and small-sample evaluation uncertainty [27] motivate explicit sampling and selection policies. Multi-dataset classifier comparisons concern an additional inference level [8]. Group-aware validation [21], clustered bootstrap [9], and unequal-cluster-size power calculations [14] address dependence under distinct designs. These sources motivate separate controls for splitting, fitting, and inference; they do not determine the direction of our recall effects.

Existing work controls shots, ways, representations, or signer splits, but these axes do not determine the within-class contribution vector across acquisition groups. Our matched-marginal design isolates this allocation axis and shows why q belongs in the reporting schema for support-set comparisons.

III. Controlled Support Allocation

A. Population, features, and classifiers

The structural scan covers 557,418 target-class records in 18 command classes across the official train, validation, and test splits. A qualifying record contains one target command, a corresponding finite 21×2 landmark slot, and wrist-centered non-wrist radial RMS above 10^{-8}. Within a class and group, identical coordinate records retain the first in archive insertion order. Every eligible training group has at least five distinct valid records. Across classes there are 796–986 eligible groups. Validation and test queries each contain 1,800 records: one hash-selected record from each of 100 groups per class. A group appearing in multiple classes remains one cluster for inference.

XY features translate landmarks by the wrist, scale by RMS, flatten to 42 dimensions, and apply ℓ_2 normalization. Mixed features concatenate XY, 20 unit bone directions (40 dimensions), and all 210 pairwise distances. Blocks are normalized separately and the concatenation divided by $\sqrt{3}$, giving 292 dimensions. A zero-length bone has direction $(0, 0)$. Neither rotation alignment nor mirror augmentation is used.

Euclidean prototypes classify by squared distance to unnormalized support class means. The extension also evaluates cosine prototypes by normalizing each mean; zero means receive zero scores. Ridge [1] regresses one-hot targets with an unpenalized intercept and predicts the largest class score. Its objective is a summed squared error plus $\lambda\|W\|_F^2$. All seven-partition ridge results use the original $\lambda = 10^{-3}$. All representations and classifiers share support IDs within each draw and condition.

B. Matching record marginals across seven partitions

For a class, let M be the number of eligible groups and $n_u \geq 5$ the size of group u. An allocation $q = (q_1, \ldots, q_U)$ has positive integer entries summing to $K = 5$. We draw an ordered list of distinct groups uniformly and take q_j records without replacement from the group in position j. Positions are randomized rather than assigned by group size. Salted SHA256 orderings implement both permutations. The seven

partitions and their concentration values are

$$q \in \{(5), (4,1), (3,2), (3,1,1), (2,2,1), (2,1,1,1), (1,1,1,1,1)\}. \quad (1)$$

For concentration $H = \sum_j (q_j/K)^2$, their respective values are 1, .68, .52, .44, .36, .28, and .20. For every partition, a particular record has inclusion probability

$$\Pr\{(u,i) \in S\} = \frac{K}{Mn_u}, \qquad \mathbb{E}[\hat{\mu}_q] = \frac{1}{M}\sum_u \mu_u, \quad (2)$$

where μ_u is the feature mean in group u. Thus changing q changes the joint support distribution while preserving the expected prototype. Equal expectation does not imply equal recall.

The original experiment uses draws 0–99; the extension uses new draws 20000–20099 on the unchanged query sets. Its development supports use draws 10020–10039. All draws use seed 20261016 and shared orderings across partitions. An exact replay reproduces original endpoint support IDs and predictions. New draw IDs resample the same training pool and reuse the same test queries; they provide no independent external test set.

C. Dispersion and uncertainty

Let B be the sample covariance of the M group means, and S_u the within-group sample covariance, each with denominator one less than its population count. Define $\bar{S} = M^{-1}\sum_u S_u$ and $C = M^{-1}\sum_u S_u/n_u$. Applying total covariance and finite-population sampling gives

$$\mathrm{Cov}(\hat{\mu}_q) = (H - 1/M)B + \bar{S}/K - HC. \quad (3)$$

This is a sampling identity, used here to predict feature dispersion from training moments. It is not a recognition-rate model. Its dependence on H is through $B - C$, so reduced concentration does not universally guarantee reduced covariance. The supplement gives the derivation and exhaustive finite-population check.

Our metric averages query correctness within each class over acquisition groups, then equally over the 18 classes. We report mean recall and sample support SD over 100 draws. For mean uncertainty, paired bootstrap replicates jointly resample support draw IDs and the global set of query-group IDs. Group multiplicities are shared across classes and configurations, then normalized within class before averaging. Replicates missing a class are redrawn. Original intervals use 1,000 replicates; extension intervals use 10,000. Neither interval captures uncertainty across datasets or landmark extractors.

The extension's focal family contains four ridge contrasts: $(3,2) - (4,1)$ and $(2,2,1) - (3,1,1)$ for XY and mixed. Besides marginal percentile intervals, we report approximate simultaneous 95% intervals. Their shared half-width is the 95th percentile of the maximum absolute centered error over the four jointly bootstrapped contrasts. One percentage point is a project-specific practical reference, set in the extension plan, rather than a universal importance threshold. Support SD describes variability of one fitted classifier; bootstrap intervals describe uncertainty in the reported mean.

TABLE I

ORIGINAL TEST RECALL (%), MEAN (SUPPORT SD), DRAWS 0–99. $K = 1$ USES THE FIRST RECORD OF THE CORRESPONDING ONE-GROUP SUPPORT AND RETAINS THE FIVE-SHOT REGULARIZER.

	$K1, U1$	$K5, U1$	$K5, U5$
XY Euclidean	37.64 (2.87)	37.88 (2.91)	37.70 (2.58)
Mixed Euclidean	45.25 (2.53)	48.50 (3.03)	55.63 (2.96)
XY ridge	43.08 (3.38)	56.12 (3.83)	68.17 (2.36)
Mixed ridge	44.71 (3.50)	64.69 (4.43)	84.78 (2.20)

TABLE II

POST-TEST EXTENSION FOCAL GAINS (PP). RIDGE AND SVM ESTIMATES USE THE SAME FIXED QUERIES AND SUPPORT DRAWS; SIMULTANEOUS INTERVALS ARE REPORTED IN THE TEXT AND SUPPLEMENT.

Redistribution	Feature	Ridge	SVM
$4+1 \to 3+2$	XY	1.9272	2.2622
	Mixed	1.7333	1.7444
$3+1+1 \to 2+2+1$	XY	0.9922	1.3317
	Mixed	0.5900	0.9622

IV. RESULTS

A. Primary pre-exposure endpoint

The retained primary comparison (Table I) changes $U = 1$ into $U = 5$ at $K = 5$, with shared $\lambda = 10^{-3}$ selected on the original validation development grid. Define the ridge representation interaction

$$T = (A_{\mathrm{mixed},U5} - A_{\mathrm{mixed},U1}) - (A_{\mathrm{XY},U5} - A_{\mathrm{XY},U1}). \quad (4)$$

The original T is 8.05 percentage points (pp), 95% interval [6.94, 9.00]. Endpoint gains are 12.05 [11.08, 13.04] for XY ridge and 20.10 [19.06, 21.14] for mixed ridge. Euclidean prototype gains are -0.18 [-1.04, 0.67] and 7.14 [6.30, 8.01]. Five further balanced query subsets retain positive ridge interactions of 7.62–8.42 pp using the same supports and regularizer.

B. Post-test allocation extension on the same population

Figure 2 compares the complete ridge and SVM allocation curves. Table II gives the four focal contrasts. The two-group gains are 1.93 pp for XY and 1.73 pp for mixed ridge, with simultaneous lower bounds of 1.33 and 1.14 pp. Within three groups, the XY gain is 0.99 pp and the mixed gain is 0.59 pp; the mixed simultaneous interval crosses zero. The allocation effect is therefore clearest for the two-group support.

The extension endpoint ridge interaction is 8.35 pp [7.36, 9.36], with XY and mixed gains of 11.00 and 19.35 pp. The post-test RBF SVM [7] cross-rule analysis selects shared $C = 100$, $\gamma = 1$ on validation; C is at the grid boundary. Its $4 + 1 \to 3 + 2$ gains are 2.26 pp [1.63, 2.90] for XY and 1.74 pp [1.11, 2.38] for mixed, using simultaneous intervals for a separate four-contrast SVM family. Three-group gains are 1.33 and 0.96 pp. The fixed-group-count allocation effect appears under both rules, while the strong

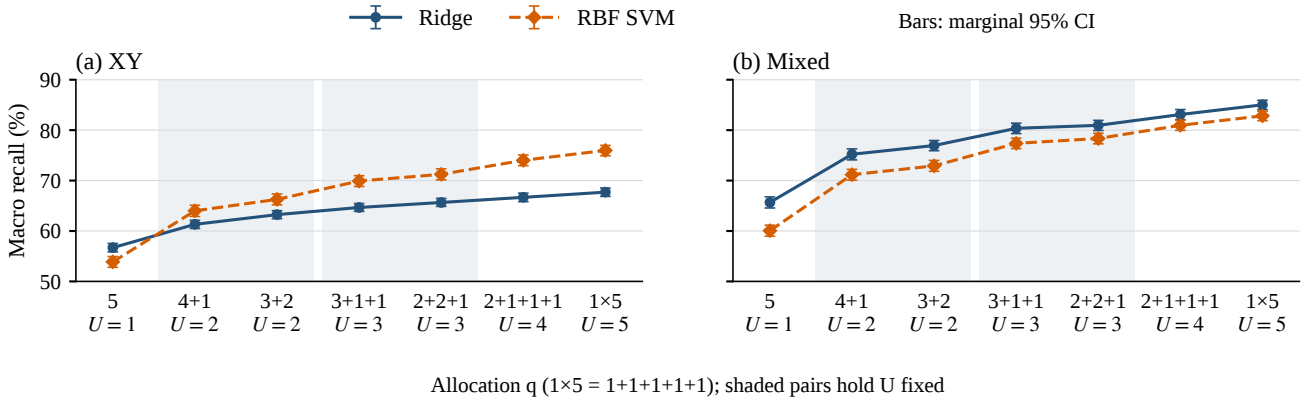


Fig. 2. All seven five-shot allocations for ridge ($\lambda = 10^{-3}$) and the validation-selected RBF SVM, using the same 100 support draws and fixed queries. Panels compare XY and mixed features. The paired $4+1$ versus $3+2$ and $3+1+1$ versus $2+2+1$ allocations hold group count fixed. Markers show means and bars show marginal paired-bootstrap 95% intervals; fixed-U simultaneous intervals are reported in the text and supplement.

ridge representation interaction is unresolved for SVM. All seven allocations are also retained for both prototype rules. XY Euclidean recall falls from 37.40% at $4+1$ to 36.33% at $3+2$, despite reduced dispersion.

C. Allocation mismatch as a reporting consequence

The direct allocation effects also determine how model gaps should be reported. For XY SVM versus XY ridge, the matched $4+1$ gap is 2.69 pp [1.65, 3.69], whereas using $4+1$ for SVM and $3+2$ for ridge reduces it to 0.76 pp [−0.34, 1.83]. This shift equals ridge's allocation effect with the opposite sign, rather than new evidence of a classifier interaction. The complete post-test mismatch analysis is provided in the supplement; models compared within one condition should share q.

D. Mechanism boundary: dispersion is not correctness

Across all seven allocations, both features show declining empirical covariance trace. Training-moment predictions lie within the corresponding support Monte Carlo intervals (complete values in the supplement). These intervals quantify finite-draw dispersion estimation, separately from the query bootstrap.

The covariance result leaves the recognition boundary explicit. XY Euclidean prototypes change by 0.04 pp [−0.86, 0.95] between extension endpoints, while mixed prototypes gain 7.24 pp [6.29, 8.21]. Cosine normalization yields −0.87 pp [−1.78, 0.07] for XY and 6.91 pp [5.96, 7.88] for mixed; no zero-mean prototype is encountered. Neither prototype rule yields a resolved XY endpoint gain, although neither has a universal zero-effect guarantee. Prototype stability and classification utility are different measured outcomes.

E. Regularization changes the interaction

The endpoint grid spans 10^{-5} to 10 in decade steps. New training development draws and the original validation queries select global, per-representation, or endpoint-specific sharing policies, with ties favoring larger λ. All selected values are interior to the grid; their test interactions span 8.10–8.35 pp. The supplement retains every parameter choice, endpoint mean, and interval.

The full grid shows $T = -0.85$ pp [−2.22, 0.54] at $\lambda = 10^{-5}$, compared with 10.46 pp [9.42, 11.49] at 10^{-2}. Validation-selected policies preserve the interaction here; arbitrary regularization does not. This grid evaluates endpoints, so fixed-U effects remain conditional on the original 10^{-3} classifier.

V. Interpretation and Reporting

Support construction should report K, U, q, eligible groups, within-group sampling, and query weighting. Models compared within one support-set condition should share the same allocation. If allocations differ, the performance gap is conditional on that mismatch and should not be interpreted directly as a classifier interaction.

The inference boundary is the fixed training pool, released landmark pipeline, and 18 known classes. Acquisition groups can combine morphology, pose, device, and collection procedure, which the metadata cannot disentangle. The valid-landmark, label-assisted protocol measures recognition conditional on target-hand localization, and the query subset is class- and group-balanced rather than naturally frequency weighted.

The allocation and SVM extensions reuse test-exposed queries and do not provide external validation. An internally pre-specified same-archive new-class study stopped before test prediction after detecting two raw train–test user overlaps; it supplies no accuracy or positive validation evidence. The evaluated support/query cohorts contain no exact cross-split coordinate pairs, although the archive contains a known pair that was not selected. Full chronology, audit scope, and saved-output replay limits are reported in the supplement.

VI. Conclusion

Shots and acquisition-group counts do not fully specify a low-shot support set. Under matched record marginals, changing the within-class allocation vector q changes recognition recall at fixed K and U: the two-group $4+1 \to 3+2$

contrasts improve both ridge representations, while three-group effects are smaller and prototype behavior differs. Dispersion and classifier controls show that support stability and recognition utility are distinct outcomes. Support-set comparisons should therefore report and match K, U, and q, together with eligible groups, within-group sampling, query weights, and selection policy.

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